

EMBR3 Air Quality Monitoring System

Technical Overview — Developer & Operator Reference

Background

The **EMBR3 Air Quality Monitoring System** is a full-stack web application built for **EMB Region 3 (Environmental Management Bureau – Central Luzon, Philippines)**. It ingests hourly particulate matter readings from four **Continuous Ambient Air Quality Monitoring Stations (AQMS)**, computes Philippine-standard AQI scores, and exposes that data through a REST API consumed by a React front-end.

The system serves three rendering contexts: a general-public interactive dashboard, a full-screen kiosk carousel, and a portrait-format LED wall display purpose-built for NLEX signage hardware. Each context is independently configurable via a PIN-gated admin panel with server-side settings persistence.

Monitoring Stations

Station	Province	Coordinates	Pollutants
Clark AQMS	Clark Freeport Zone, Pampanga	15.177°N, 120.536°E	PM10
San Fernando AQMS	San Fernando, Pampanga	15.056°N, 120.644°E	PM10
Meycauayan AQMS	Meycauayan, Bulacan	14.728°N, 120.958°E	PM10, PM2.5
Zambales AQMS	Santa Cruz, Zambales	15.775°N, 119.915°E	PM10, PM2.5

Tech Stack Summary

Layer	Technologies
Frontend	React 19, Vite 7, Ant Design 5, Recharts 3, Tailwind CSS, globe.gl, Three.js
Backend	Node.js 18+, Express 5, MongoDB Atlas, ExcelJS, Nodemailer
External APIs	Open-Meteo, OpenWeatherMap (tile proxy), Nominatim, Google Sheets CSV, Windy.com, IQEarth
Infrastructure	Hostinger KVM2 VPS, Nginx, PM2, Cloudflare CDN

Features

1. Live AQI Dashboard (/)

- **AQI Hero Card** — animated sky background driven by real-time Open-Meteo weather code and cloud cover; day/night aware; supports 7+ weather scenarios (clear, clouds, rain, thunder, snow, fog, overcast).
- **Station selector** — dropdown persists station selection via AES/HMAC-encrypted `localStorage`.
- **Dual-pollutant rendering** — merged stations (Meycauayan, Zambales) render two side-by-side gauges and two `AqiHeroCard` instances from a single API response.
- **AqiCategoryMeter** — SVG arc gauge with gradient fill, animated on value change.
- **HistoricalAqiGraph** — Recharts line/bar chart showing 24-hour rolling averages and daily AQI trends.
- **AqiCalendar** — Day-level heatmap with click-through modal for hourly breakdown.
- **WindMapCard** — Windy.com iframe embed pinned to station coordinates.
- **Stale-data watermark** — Overlay rendered if latest reading timestamp is >7 days old.

2. Kiosk Mode (/embr3-latestaqi , /)

- **Auto-cycling carousel** — alphabetically sorted station list; configurable dwell time (default 25 sec).
- **Warmup prefetch** — sequential `useEffect` chain fetches all station data on mount to pre-warm caches; uses time-gated promises to prevent network flooding.
- **YouTube embed section** — agency messaging loop via `<iframe>` with `autoplay=1&mute=1&loop=1`.
- **Maintenance overlay** — `MaintenanceOverlay` component injected when `kioskSettings.maintenance.enabled` is `true`.

3. Kiosk + ARTA Variant (/with-arta)

- Same carousel logic as standard kiosk with an interstitial `ARTA` video loop injected every N station rotations.
- Route-gated — `App.jsx` uses `React.lazy()` for code splitting; ARTA variant is a separate lazy chunk.

4. NLEX LED Wall (/nlex , /nlex-preview)

- **Native fallback** — `GET /nlex` serves a lightweight ES5 inline HTML response (`NlexLedWall` native route in Express) for VNNOX signage browser compatibility; no React bundle.

- **Browser preview** — `/nlex-preview` or `?mode=browser` loads the full React `NlexLedWall` component.
- **Dimensions** — 960×1536 px portrait viewport.
- **Grid mode** — 2×2 CSS Grid, all four stations simultaneously.
- **Carousel mode** — absolute-positioned fullscreen card per station, `z-index` cycling driven by `setInterval`.
- **SVG Gauge** — custom gauge: 270° arc, gradient stroke, animated pointer, configurable size (330 px dual / 560 px solo).
- **Spotlight mode** — featured station rendered with expanded description overlay and distinct background treatment.
- **BroadcastChannel sync** — `NlexLedWall` subscribes to `nlex-settings-update` channel; settings changes in `/admin/nlex-settings` propagate instantly via `postMessage` without page reload.
- **Clock granularity** — 1-sec interval in browser preview; 60-sec interval in native mode for VNNOX compat.

5. NLEX Admin Settings (`/admin/nlex-settings`)

- Tabs-based UI using Ant Design `<Tabs>` and `<Form>`.
- **Settings schema** — persisted to MongoDB `nlex_settings` collection; served via `GET /api/nlex-settings`.
- **Draft/save pattern** — local React state mirrors server state; mutations staged until user confirms with "Save Changes" (`PUT /api/nlex-settings`).
- **Cross-tab propagation** — on save, broadcasts `nlex-settings-update` event to all open tabs via `BroadcastChannel`.
- Configurable: station visibility, display mode, per-station carousel duration, gauge visibility, spotlight station, theme override (auto/light/dark), AQI description text per band, component visibility (header, subtitle, legend, footer).

6. Kiosk Settings (`/admin/kiosk-settings`)

- Same draft/save/broadcast pattern as NLEX settings.
- Persisted to MongoDB `kiosk_settings` collection.
- Configurable: per-station AQI value/timestamp visibility, section toggles (weather, hourly, wind map, carousel, YouTube, contact), cycle interval, maintenance mode (enabled + custom message).

7. Map Page (`/admin/map`)

- **globe.gl** 3D globe with `GlobeGL` component; basemap toggle (satellite/streets/dark) via tile URL swap.
- **Google Maps embed** flat map fallback.
- **Sequential weather fetch** — one station per tick (500 ms gap) using a recursive `setTimeout` chain; 5-min refresh interval.
- **Station detail modals** — photo, coordinates, current conditions from `useStationWeather` hook.

8. Charts & Tabular Data (`/admin/charts` , `/admin/tabular/:province`)

- **Recharts** bar and line charts rendered from `/api/viz-data` and `/api/pm10-data`.
- **Ant Design** `<Table>` with server-side filtered dataset from `/api/tabular/:province/:pollutant`.
- **CSV export** and **email share** (`POST /api/share-email`).
- **Export logging** — every download/email writes to MongoDB `export_logs` collection.

9. Admin Auth (`AdminPinGate`)

- Wraps all `/admin/*` routes in a PIN challenge component.
- On success, `POST /api/admin/verify-pin` returns a signed JWT; stored in AES-encrypted `sessionStorage`.
- Subsequent admin API calls include `Authorization: Bearer <token>` header.

Functions

AQI Calculation (`server/services/aqiCalculator.js`)

- Implements **DENR DAO 2000-81 / NAAQGV breakpoints** for PM10 and PM2.5.
- Uses linear interpolation: $AQI = ((AQI_{High} - AQI_{Low}) / (BP_{High} - BP_{Low})) * (Cp - BP_{Low}) + AQI_{Low}$
- Separate breakpoint tables for PM10 ($\mu g/m^3$ 24h avg) and PM2.5 ($\mu g/m^3$ 24h avg).
- Returns `{ aqi, status, category, color }` per reading.
- **24-hour rolling average** — sliding window computed from hourly rows before calling the breakpoint function.

Data Ingestion Pipeline

1. **Google Sheets CSV** — `server/services/googleSheets.js` fetches CSV exports from published Google Sheets (one URL per station/pollutant, configured in `server/config/sheets.js`).
2. **CSV Parsing** — auto-detects date, concentration, and pollutant columns; maps to normalized row schema.
3. **AQI Enrichment** — parsed rows passed through `aqiCalculator`; rolling averages attached.
4. **MongoDB Upsert** — enriched rows upserted into per-collection documents keyed by `province + pollutant`.
5. **Scheduled refresh** — `setInterval` at 15-min cadence; startup sync on server boot forces full refresh.
6. **Fallback chain** — MongoDB cache → Google Sheets live fetch → Excel workbook (ExcelJS).

Weather Integration (`server/routes/station.js` , `front-end/src/hooks/useStationWeather.js`)

- **Open-Meteo API** (free, no auth) — GET `https://api.open-meteo.com/v1/forecast?latitude=X&longitude=Y&...`
- Fetches: `temperature_2m`, `relative_humidity_2m`, `surface_pressure`, `wind_speed_10m`, `wind_direction_10m`, `weathercode`, `uv_index`, `cloudcover`, `apparent_temperature`, plus 24h hourly forecast.
- Weather codes mapped to human-readable labels and background gradient keys.
- **Reverse geocoding** — Nominatim GET `https://nominatim.openstreetmap.org/reverse?lat=X&lon=Y` for place name display.

Encrypted Storage (`front-end/src/util/secureStorage.js`)

- All `localStorage` / `sessionStorage` writes pass through AES-256-CBC encryption.
- **HMAC-SHA256** integrity tag appended; read operations verify tag before decrypting.
- Encryption key derived from a deployment-time secret (`VITE_STORAGE_SECRET`); fallback to a deterministic app-scoped key.
- Used for: admin token, station selection, NLEX settings cache, kiosk settings cache, NLEX bundle cache.

Email Service (`server/services/emailService.js` , `server/routes/email.js`)

- **Nodemailer + Gmail SMTP** — transporter configured via `EMAIL_USER` / `EMAIL_APP_PASSWORD` env vars.
- `POST /api/share-email` accepts: `{ to, province, pollutant, data[], filters }`.
- Builds HTML table from filtered dataset rows; sends via `transporter.sendMail()`.
- Export event logged to MongoDB after successful send.

Excel Workbook Fallback (`server/services/workbook.js`)

- **ExcelJS** reads `.xlsx` workbook from local path or SharePoint URL.
- Provides `GET /api/viz-data` and `GET /api/pm10-data` endpoints as offline data sources.
- `GET /api/viz-data/diagnostics` returns row count, column headers, and sheet names for debugging.

Proxy & Caching (`server/routes/proxy.js`)

- **OWM tile proxy** — `GET /api/tiles/owm/:layer/:z/:x/:y.png` fetches OpenWeatherMap tile, buffers response, returns with `Cache-Control: max-age=300`.
- Shields the client-side map from direct API key exposure; key stored server-side in `OWM_API_KEY` env var.

Advantages

Multi-Tier Data Resilience

- Three-layer fallback (MongoDB → Google Sheets → Excel) ensures the API never returns empty-handed; each layer is independently refreshable.
- Startup sync guarantees post-deploy freshness without manual cache invalidation.

Encrypted Browser Storage

- AES/HMAC-hardened `localStorage` prevents client-side tampering with admin tokens and display configuration.
- Automatic transparent encryption/decryption means no component code needs to handle raw keys.

Independent Display Mode Architecture

- Dashboard, Kiosk, and NLEX LED wall share a single REST API but maintain completely independent settings schemas, persistence keys, and rendering logic.
- New display modes can be added as new React page components and a new MongoDB settings collection without modifying existing modes.

Real-Time Admin-to-Display Sync

- `BroadcastChannel` pattern propagates settings changes to all open display tabs/windows in the same browser session instantaneously — no polling required.
- MongoDB persistence ensures changes survive across sessions and devices.

Dual-Pollutant Abstraction

- Meycauayan and Zambales stations report both PM10 and PM2.5. The API serves these as separate records; the front-end `stations.js` config defines a `merged` flag that triggers dual-gauge rendering in all display modes transparently.
- Existing single-pollutant display components require zero modification.

Security Posture

- **JWT admin auth** — short-lived tokens; no persistent session cookies.
- **CORS whitelist** — origins restricted per environment via `CORS_ORIGIN` env var.
- **Rate limiting** — per-IP in-memory rate limiter (default 600 req / 60 sec) on all API routes.
- **Security headers** — `helmet` -equivalent headers: CSP, X-Frame-Options (with NLEX iframe carve-out), HSTS, X-Content-Type-Options, Referrer-Policy.
- **No API key exposure** — OWM tiles proxied server-side; Google Sheets accessed via published CSV (no OAuth token in client).

Lazy Loading & Code Splitting

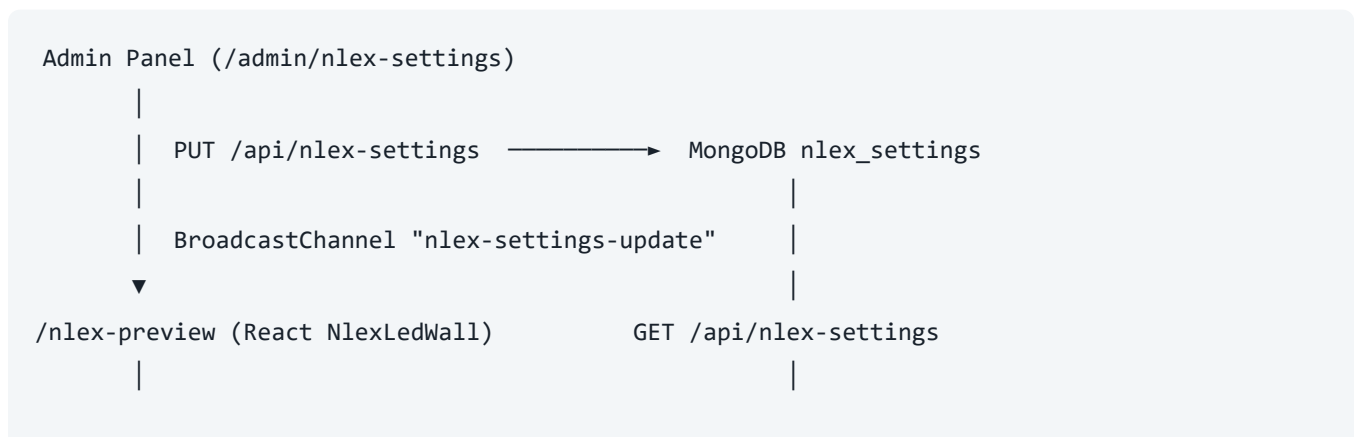
- `React.lazy()` + `Suspense` for all route-level components; Vite generates per-route chunks.
- Admin routes, NLEX LED wall, and ARTA variant are separate bundles not loaded on initial dashboard paint.

NLEX Featured Overview

Purpose

The NLEX integration delivers live air quality data from EMB Region 3 onto large-format LED signage boards operated by NLEX along the North Luzon Expressway. The goal is to give motorists ambient awareness of regional air quality without requiring them to interact with a device.

Architecture



		(same React component)	▼
/nlex		(VNNOX: served as ES5 inline HTML)	/nlex auto-reload

Native Fallback (/nlex route)

The VNNOX LED wall browser is a Chromium-based embedded browser with limited ES6+ support. The Express server's /nlex route handler detects the native request and responds with a hand-crafted ES5 HTML string (NlexLedWall in server/routes/station.js or dedicated route) that:

- Embeds all CSS inline (no external stylesheets)
- Uses XMLHttpRequest instead of fetch
- Avoids ES6 modules, arrow functions in outer scope, and const / let block declarations where not supported
- Polls /api/aqi/latest and /api/nlex-settings on a 60-second interval
- Renders station cards with inline SVG gauges

React Component (NlexLedWall.jsx)

Used on /nlex-preview and for desktop testing:

- Reads settings from NlexSettingsContext (hydrated from GET /api/nlex-settings on mount)
- Subscribes to BroadcastChannel for real-time config updates
- Manages carousel activeIndex state via useRef -tracked setInterval
- Per-station duration from settings.carousel.durations[stationKey] (seconds → ms)
- Animated weather backgrounds resolved from Open-Meteo weathercode + is_day flag

Settings Schema (MongoDB nlex_settings)

```
{
  "displayMode": "grid | carousel",
  "stations": {
    "clark": { "visible": true },
    "sanFernando": { "visible": true },
    "meycauayan": { "visible": true },
    "zambales": { "visible": true }
  },
  "carousel": {
    "durations": {
      "clark": 10,
      "sanFernando": 10,
      "meycauayan": 10,

```



```

    "zambales": 10
  }
},
"gauge": { "visible": true },
"spotlight": { "enabled": false, "station": "clark" },
"theme": "auto | light | dark",
"aqiDescriptions": {
  "good": "...",
  "fair": "...",
  "unhealthySensitive": "...",
  "veryUnhealthy": "...",
  "acutelyUnhealthy": "...",
  "emergency": "..."
},
"components": {
  "header": true,
  "datetime": true,
  "subtitle": true,
  "legend": true,
  "footer": true
}
}

```

LED Wall Rendering Specifics

- **Viewport** — forced to 960×1536 px via CSS `width / height` on root container; no viewport meta scaling.
- **Font sizes** — station name: 34 px; AQI value: 96 px; category label: 28 px; all sizes set in `px` (not `rem`) for predictable LED rendering.
- **Gauge dimensions** — 330 px (dual-pollutant layout) / 560 px (single-pollutant layout).
- **Dual-pollutant layout** — Meycauayan and Zambales render a two-column flex row with PM10 on the left and PM2.5 on the right inside a single station card.
- **AQI scale legend** — horizontal color bar pinned to card bottom showing all 6 categories with current value indicator.
- **Clock update** — 60-second `setInterval` in native mode (VNNOX compat); 1-second in React preview.

API Endpoints Used by NLEX Display

Method	Endpoint	Purpose
GET	/api/aqi/latest	Fetch latest readings for all stations
GET	/api/nlex-settings	Fetch current display configuration
PUT	/api/nlex-settings	Update configuration (admin, JWT-protected)
GET	/api/station/current	Fetch current weather (for background)

API Reference (All Endpoints)

Method	Path	Auth	Description
GET	/	—	Service status
GET	/health	—	Health check + timestamp
GET	/api/aqi/latest	—	Latest AQI for all stations
GET	/api/aqi/last-days	—	Rolling N-day AQI history
GET	/api/tabular/:province/:pollutant	—	Full tabular dataset with rolling averages
GET	/api/backup/status	—	Dataset freshness info
GET	/api/backup/check/:province/:pollutant	—	Check if updates available
POST	/api/backup/sync	JWT	Force manual backup cycle
POST	/api/export-log	—	Log export event
GET	/api/export-logs	JWT	Recent export log entries
GET	/api/station/meta	—	Station metadata (name, coords)
GET	/api/station/current	—	Current weather conditions
GET	/api/station/forecast	—	N-day daily forecast
GET	/api/viz-data	—	Viz time-series from workbook
GET	/api/pm10-data	—	PM10 time-series from workbook
GET	/api/viz-data/diagnostics	—	Workbook diagnostics

Method	Path	Auth	Description
POST	/api/share-email	—	Send HTML report by email
GET	/api/tiles/owm/:layer/:z/:x/:y.png	—	Proxied OWM map tile
GET	/api/reverse-geocode	—	Nominatim reverse geocode
POST	/api/admin/verify-pin	—	Verify PIN, return JWT
GET	/api/nlex-settings	—	Public NLEX settings
PUT	/api/nlex-settings	JWT	Update NLEX settings
GET	/api/kiosk-settings	—	Public kiosk settings
PUT	/api/kiosk-settings	JWT	Update kiosk settings

Environment Variables

Variable	Location	Purpose
MONGO_URI	server	MongoDB Atlas connection string
EMAIL_USER	server	Gmail account for email sending
EMAIL_APP_PASSWORD	server	Gmail app password
ADMIN_PIN	server	Hashed admin PIN for JWT issuance
JWT_SECRET	server	Secret for signing admin JWTs
OWM_API_KEY	server	OpenWeatherMap API key (tile proxy)
CORS_ORIGIN	server	Allowed CORS origin(s)
PORT	server	Express server port (default 3000)
VITE_API_BASE_URL	front-end	Backend API base URL
VITE_STORAGE_SECRET	front-end	Key for AES-encrypted localStorage
VITE_STATION_*_LAT/LON	front-end	Per-station coordinate overrides